

Selectime experiment guideline

1 Question and scope

Test whether validation can combine frozen foundation-model input scopes and retrieved covariates on Adapte’s 90 dataset/frequency/term tasks with its four dataset exclusions. Chronos-2 reports univariate and multivariate baselines, task and item/variant scope selectors, task and item/variant scope mixtures, fixed-alpha scope ridge, and top- K mixtures for $K \in \{1, 5, 10, 15, 20\}$. TS-ICL reports univariate and the five top- K mixtures. Chronos-Bolt reports univariate only. Raw top- K forecasts are intermediate mixture inputs. Self-augmentation, broad selectors over all candidates, and direct-horizon transfer are excluded.

2 Windows and retrieval

The official TIME test length and horizon H define test windows. Validation origins walk backward from the first test origin at stride H and the same phase, capped by both the requested validation length and the official test date count. The default requested validation length equals the test length. Scope ridge uses exactly these windows as fitting rows; there is no additional training interval.

A validation row is usable only when its available target history and H -step future each contain finite support. Unusable rows never reach a backbone and contribute no loss. Validation retrieval ends before validation starts; test retrieval ends before test starts. Neighbor futures are finite, wholly before the applicable boundary, and available before the query. No test observation enters either datastore.

Use instance-normalized Euclidean lookback distance with at least 80% finite overlap. Require complete maximum- K support and reuse ordered prefixes for smaller K . Rescale retrieved histories and observed futures to query units using lookback-only statistics. All-series scope is the default; same-series scope remains an explicit ablation. Datastore caps are optional and disabled by default.

The focused retrieval ablation changes one axis at a time: same-user scope, the full datastore, raw distance, or disabled query scaling. A separate oracle diagnostic refits implemented combinations on official test labels after the matching default evaluations complete. It is an upper bound and is never reported as a deployable selection rule.

3 Hard scope selection

The hard selectors compare univariate and multivariate validation forecasts by per-date MSSE. Task selection averages variates within each date; per-variate selection scores each item/variant separately. Use the paired moving-date-block bootstrap one-standard-error rule with 1000 replications by default. The automatic block length is the larger of rounded $N^{1/3}$ and $\lceil H/s_{\text{val}} \rceil$, capped at $N - 1$. Prefer univariate within the admissible set. No usable date selects univariate; one date uses the observed minimum. One run seed controls backbone stochasticity and bootstrap sampling.

4 Mixtures and scope ridge

For each win-frequency mixture, compare univariate and one alternative on eligible validation rows. Exclude alternative fallback rows and non-finite losses. With w alternative wins among n rows, counting ties as one half, freeze

$$p = \frac{1 + w}{2 + n}, \quad \hat{y} = (1 - p)\hat{y}_{\text{uni}} + p\hat{y}_{\text{alt}}.$$

No usable row gives $p = 0$. Otherwise, the mixture remains pure univariate unless there are at least 10 validation trials, validation support exceeds 10% of matching test rows, and the empirical alternative win rate exceeds 10%. Scope weights are fitted once per task or per item/variant; each top- K weight is fitted once per task.

For task-level scope ridge, let $d = \hat{y}_{\text{multi}} - \hat{y}_{\text{uni}}$ and $r = y - \hat{y}_{\text{uni}}$. For finite target cells j , use MSSE weight a_j , the reciprocal of the row’s observed-future count times its seasonal scale. With configured but unvalidated $\alpha = 1$,

$$p = \frac{\sum_j a_j d_j r_j}{\alpha + \sum_j a_j d_j^2}, \quad \hat{y} = \hat{y}_{\text{uni}} + pd.$$

Do not clip p . No usable fitting row records an unfitted vanilla fallback with $p = 0$. Freeze every choice or coefficient before test.

5 Execution and artifacts

Hydra configuration is `src/timebench/conf/experiment.yaml`; task settings are `src/timebench/config/datasets.yaml`; the entry point is `src/scripts/run_selectime.py`. The Chronos-2, TS-ICL, and Bolt launchers are respectively `scripts/submit_experiment.sh`, `scripts/submit_ts_icl.sh`, and `scripts/submit_chronos_bolt.sh`. The scheduler first prepares data, then extracts and predicts validation windows, calibrates the controls, extracts and predicts test windows, assembles the controlled forecasts, evaluates them, and builds the report. Forecast stages finalize canonical vanilla before dependent candidates.

Products live under `outputs/<surface>/scope_selection/<backbone>/`; reports remain below that experiment’s `reports/` directory, and logs use the matching `logs/<surface>/scope_selection/` hierarchy. Timestamps and launch identifiers remain in manifests and logs rather than directory names. Every stage allocates an authoritative schema-1 scientific configuration and lifecycle manifest before work; run directories do not duplicate it in a `config.json`. Fully written tasks become `computed`; a successful scheduler step finalizes them as `completed`. Exact completed work is reused and interrupted tasks restart from their beginning. Reports retain consumed manifests, metric coverage, fallbacks, hard-choice counts, mixture weights, and ridge support. Assembled methods have no independently measured inference latency.

6 Verification and evidence

The focused regression module is `src/tests/test_selectime_contract.py`. It covers chronology, retrieval, hard scope selection, task/per-variant mixtures, scope ridge, lifecycle, and report contracts without foundation inference. Run a narrow remote smoke before the complete study. Existing result documents describe earlier configurations; fresh runs are required for this reduced reported method set.